



END-USER OPERATING PROCEDURE

MariSoft-AIS Structures

Daily inspection, task, evidence, event, and anomaly workflow

Application	MariSoft-AIS Structures v11.0.0
Audience	Inspectors, engineers, reviewers, and workpack users
Document version	1.0
Effective date	17 July 2026

Controlled operational guide

1. Purpose and operating rules

This procedure explains the standard end-user workflow in MariSoft-AIS Structures: connect to the correct data source, select the inspection context, perform task work, capture supporting evidence, manage events and anomalies, and close the session safely.

Golden rule: Before adding or changing records, verify the Server, Database, Asset, Schedule, and Workpack values in the status bar.

1.1 Intended users

- Inspectors recording task progress and inspection observations.
- Engineers or reviewers checking events, anomalies, supporting media, and actions.
- Workpack users preparing, copying, exporting, or reviewing inspection data.

1.2 Before you begin

- Obtain the approved job folder and database file, or approved SQL Server connection details.
- Confirm which Region, Site, Asset, Schedule, and Workpack you must use.
- Ensure any required video folder, 3D model, point-cloud data, or documents are accessible.
- Do not create a new database, clone a workpack, delete records, or renumber anomalies unless your role authorizes it.

Access: Available buttons and pages depend on connection state, workpack data, configuration, and user role. A disabled command is not necessarily an error.

2. Interface at a glance

Use these areas to orient yourself after the application opens.

Area	What it contains	End-user purpose
Top menu	Workspace, Database, DV Connection, Theme	Save the layout, back up/export data, and choose the display theme.
Left navigation	Database Connection, Inspection, Templates	Connect first; Inspection becomes available after data is loaded.
Inspection workspace	Asset tree, task data, video, events, anomalies, models	Perform and review inspection work in context.
Status bar	Server, database, asset, schedule, workpack, NodeJS status	Confirm that the correct operating context is active before editing.

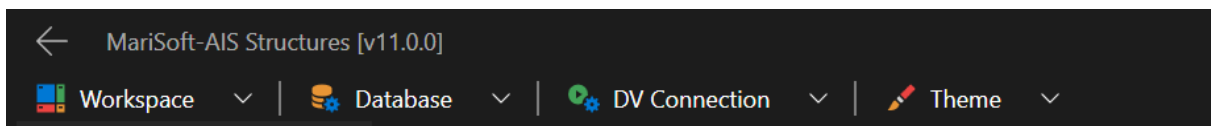


Figure 1. Main application menu and version banner

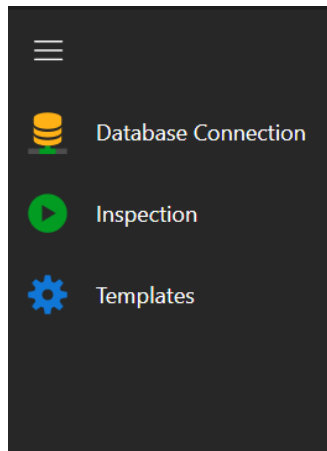


Figure 2. Primary navigation panel



Figure 3. Operating-context status bar

3. Standard operating sequence

Use this sequence for a normal inspection or review session.

- 1 Open the job folder** On Database Connection, browse to or drop the approved job folder. Use the plus button only when a new standard folder structure is required.
- 2 Connect to the data source** Choose either Database File (.db) or SQL Server, provide the required details, and select the correct database.
- 3 Select the operating context** Choose Region, Site, Asset, Schedule, and Workpack in order. Later selections depend on earlier ones.
- 4 Load the data** Select Load Data and wait for processing to finish. Confirm the Inspection page becomes available.
- 5 Verify the status bar** Check the active server/database, asset, schedule, and workpack before editing any record.
- 6 Perform or review work** Use the inspection tree and Task Data tabs to work with tasks, videos, events, anomalies, evidence, documents, and models.
- 7 Finish and safeguard** End any active task, save completed edits, back up when required, and close only after long-running export or media operations finish.

4. Connect and load inspection data

The Database Connection page collects the job folder, data-source type, credentials or database file, and the location/workpack context. The example below shows a configured operating context; always use the values approved for your own assignment.

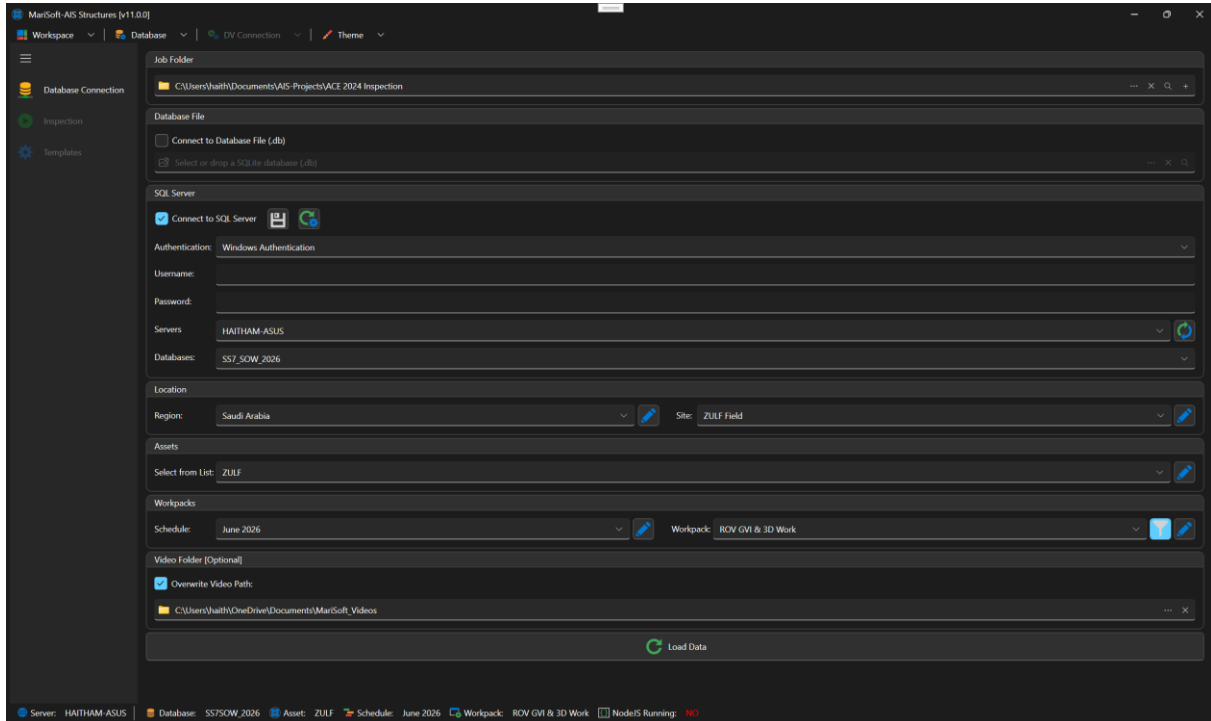


Figure 4. Database Connection page and selected workpack context

4.1 Connect to a SQLite database file

- 1 Select Database File** Tick Connect to Database File (.db).
- 2 Choose the file** Browse to or drop the approved .db file into the database field.
- 3 Select location and workpack** Choose Region, Site, Asset, Schedule, and Workpack.
- 4 Add video path if needed** Use Video Folder [Optional] when media is stored separately. Apply Overwrite Video Path only when the approved media location has changed.
- 5 Load Data** Wait until the busy indicator clears, then verify the status bar.

4.2 Connect to SQL Server

- 1 Select SQL Server** Tick Connect to SQL Server.
- 2 Choose authentication** Select the approved authentication mode. Enter username and password only when SQL credentials are required.
- 3 Choose server and database** Refresh the server list if necessary, then select the approved server and database.

4 Complete the context Select Region, Site, Asset, Schedule, and Workpack, then choose Load Data.

Security: Never share database passwords or store them in an uncontrolled document. Use the save/load settings buttons only in accordance with local IT policy.

4.3 Connection acceptance check

- Inspection navigation is enabled.
- The status bar shows the expected server/database, asset, schedule, and workpack.
- The inspection tree and Task Data area display records expected for that workpack.
- If the context is wrong, stop and reconnect before entering data.

5. Navigate the inspection workspace

The inspection workspace links the asset/component hierarchy to operational data. Select a tree node first; the task, event, anomaly, evidence, document, and model views then operate in that context.

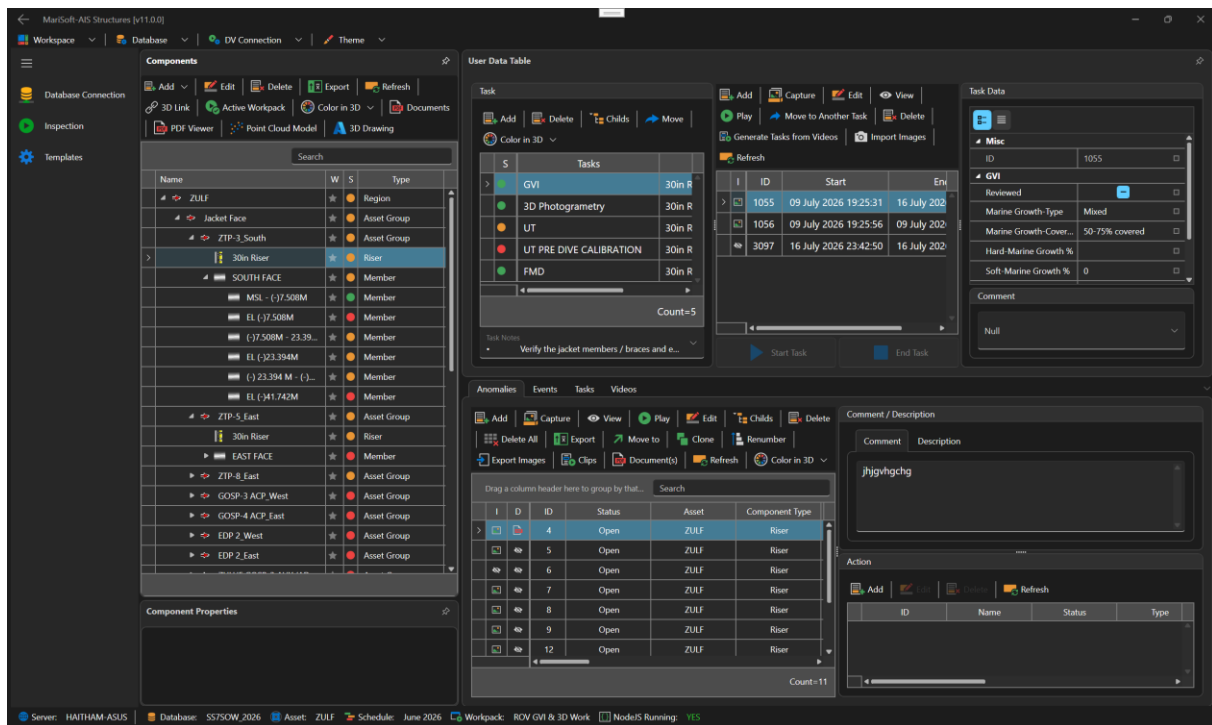


Figure 5. Inspection workspace overview

5.1 Work with the inspection tree

- Use Expand Selected Node, Collapse Selected Node, Expand All Nodes, or Collapse All Nodes to control the hierarchy.
- Select the correct asset/component node before adding or editing task-related data.
- Use Add, Edit, Move, and Delete only when maintaining the hierarchy is part of your role.

- Enable Drag and Drop only for deliberate tree maintenance; review the destination before saving tree changes.
- Use 3D Link to associate the selected tree component with selected items in the open 3D drawing.

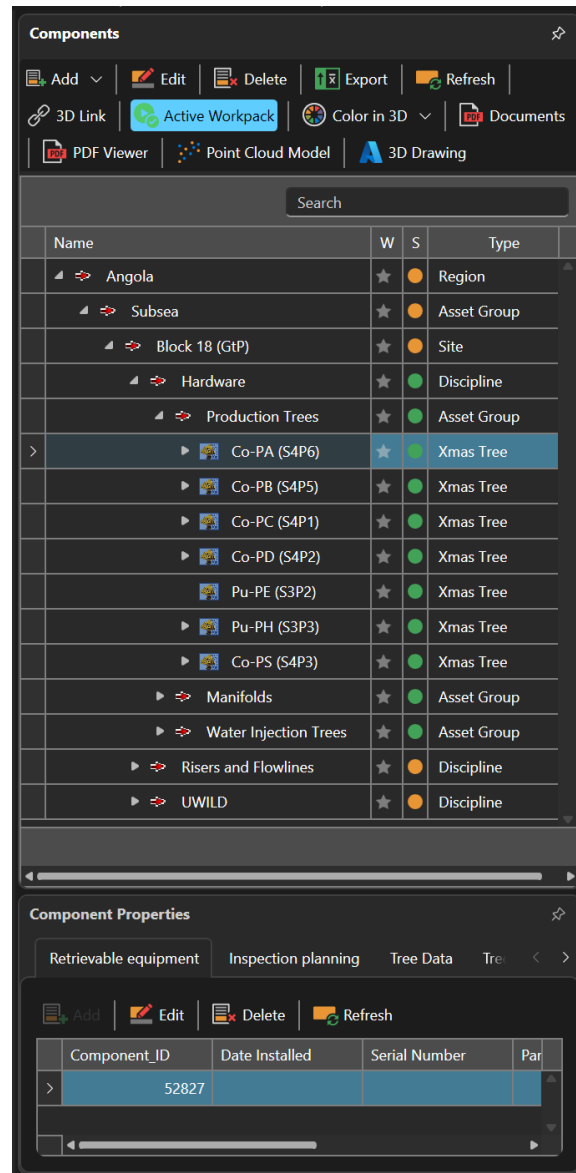


Figure 6. Component hierarchy and component properties

Caution: Move, Delete, drag-and-drop, and 3D linking can change relationships used by other users. Confirm the selected node and destination before accepting.

5.2 Use drawings, models, and documents

- Open 3D Drawing to view or select linked model items; Color in 3D highlights records using the chosen color.
- Use Point Cloud Model when point-cloud data is available for the selected context.

- Use PDF Viewer or Documents to open attached reference material.
- If a view is blank, confirm the selected component has a valid model/document link and that the source file is accessible.

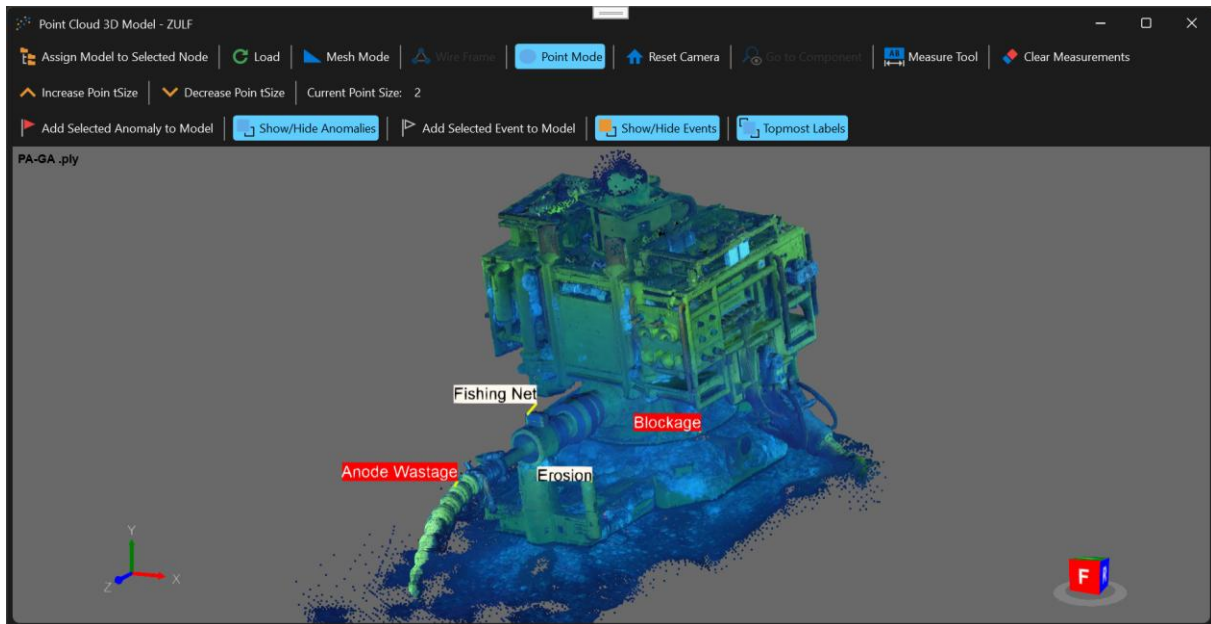


Figure 7. Point-cloud model with anomaly and event labels

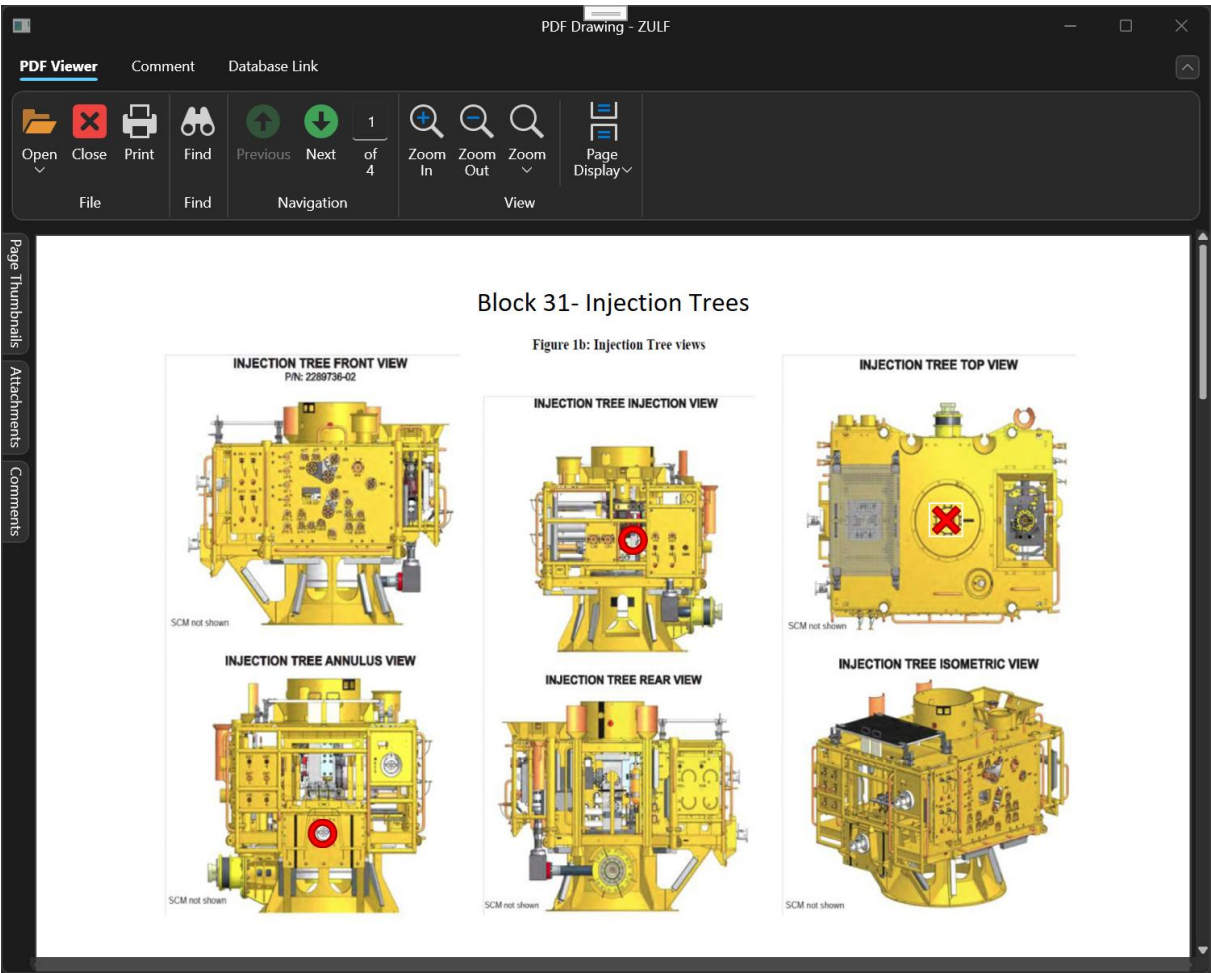


Figure 8. PDF Viewer with linked technical drawing

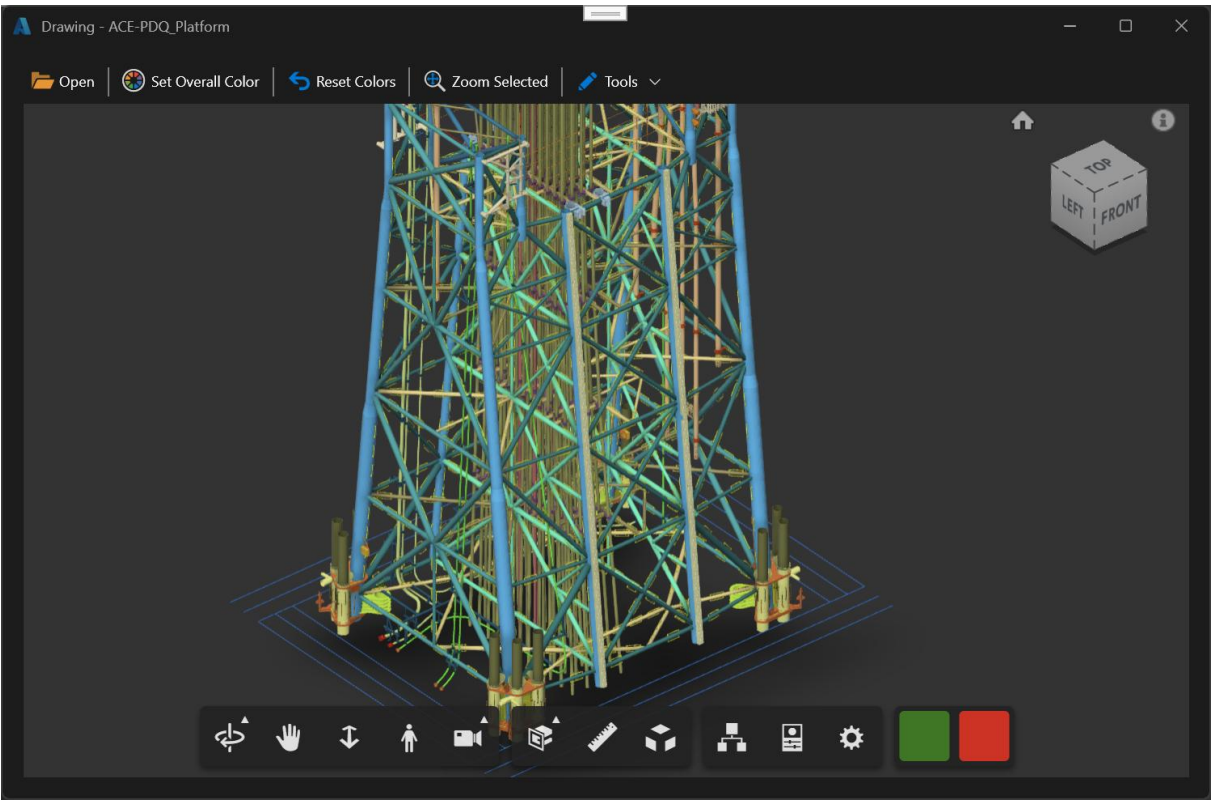


Figure 9. Interactive 3D drawing view

Model and drawing context: Confirm the title bar and selected component before assigning a model, adding labels, applying colors, measurements, or database links.

6. Perform and manage tasks

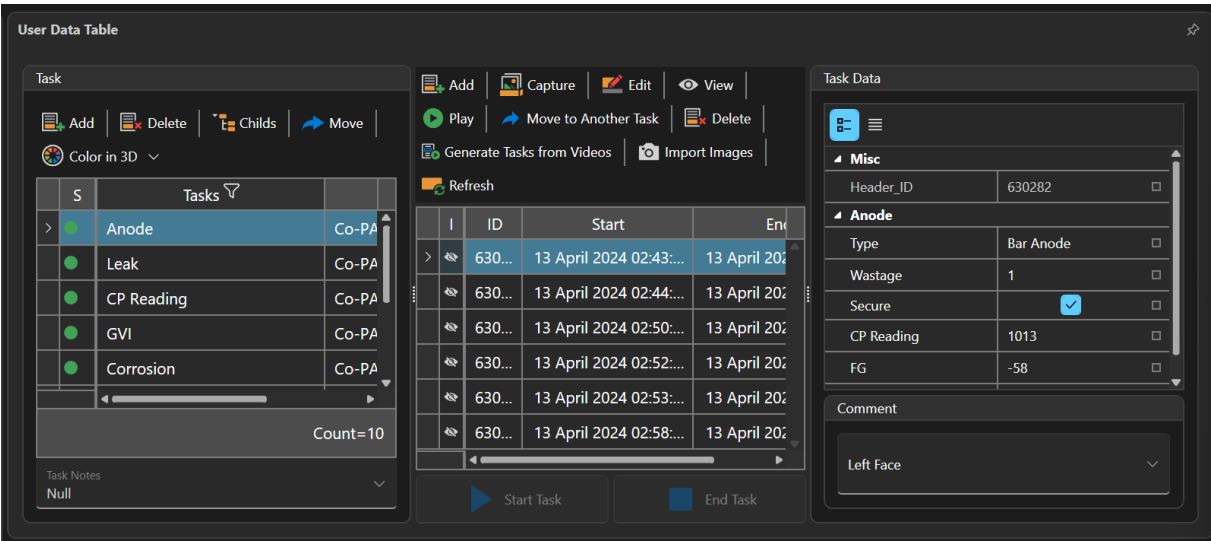


Figure 10. Task selection, task records, and task data

6.1 Start a task

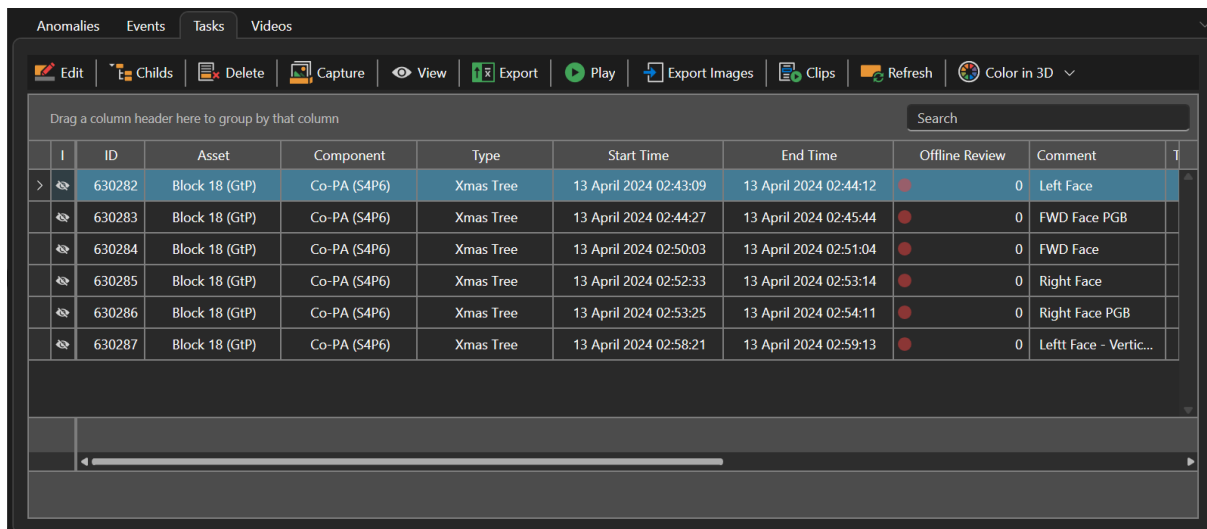
- 1 Select the component** Choose the required node in the inspection tree.
- 2 Select the task** In Task Data or Tasks, select the correct task record. Confirm its component, type, and description.
- 3 Start Task** Choose Start Task to use the current time, or Set Start Time when an approved recorded time must be entered.
- 4 Verify time and context** Check the Start Time and confirm the task remains associated with the intended component.

6.2 Record evidence and notes

- Use Capture to take a task image from the available media source.
- Use Import Images for existing approved images; review the selected files before import.
- Use Play to open linked video. Use Set time where the current video position must be stored against a record.
- Use Comment or Task Notes for concise, factual inspection notes. Avoid unsupported conclusions.
- Use Documents to attach or open supporting files when required by the workpack procedure.

6.3 End a task

- 1 Review the record** Confirm notes and evidence are attached to the correct task and component.
- 2 End Task** Choose End Task to use the current time, or Set End Time for an approved recorded time.
- 3 Check the time range** Ensure End Time is later than Start Time and is consistent with the relevant video segment.
- 4 Refresh if required** Use Refresh to confirm the saved state shown in the grid.



The screenshot shows a software interface with a top navigation bar containing 'Anomalies', 'Events', 'Tasks', and 'Videos'. Below this is a toolbar with icons for 'Edit', 'Childs', 'Delete', 'Capture', 'View', 'Export', 'Play', 'Export Images', 'Clips', 'Refresh', and 'Color in 3D'. A search bar is located on the right. The main area displays a table of task records with columns: ID, Asset, Component, Type, Start Time, End Time, Offline Review, and Comment. The table contains six rows of data, all for 'Block 18 (GtP)' and 'Co-PA (S4P6)' components, with various 'Xmas Tree' types. The 'Offline Review' column shows a red dot and the number '0' for each record. The 'Comment' column contains descriptions like 'Left Face', 'FWD Face PGB', 'FWD Face', 'Right Face', 'Right Face PGB', and 'Leftt Face - Vertic...'. Below the table is a horizontal scrollbar.

ID	Asset	Component	Type	Start Time	End Time	Offline Review	Comment
630282	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:43:09	13 April 2024 02:44:12	0	Left Face
630283	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:44:27	13 April 2024 02:45:44	0	FWD Face PGB
630284	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:50:03	13 April 2024 02:51:04	0	FWD Face
630285	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:52:33	13 April 2024 02:53:14	0	Right Face
630286	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:53:25	13 April 2024 02:54:11	0	Right Face PGB
630287	Block 18 (GtP)	Co-PA (S4P6)	Xmas Tree	13 April 2024 02:58:21	13 April 2024 02:59:13	0	Leftt Face - Vertic...

Figure 11. Completed task records and review controls

Do not leave active tasks: Before closing the application or changing workpacks, end any task that has been completed. If work is genuinely continuing, follow the local shift-handover rule.

7. Work with video records

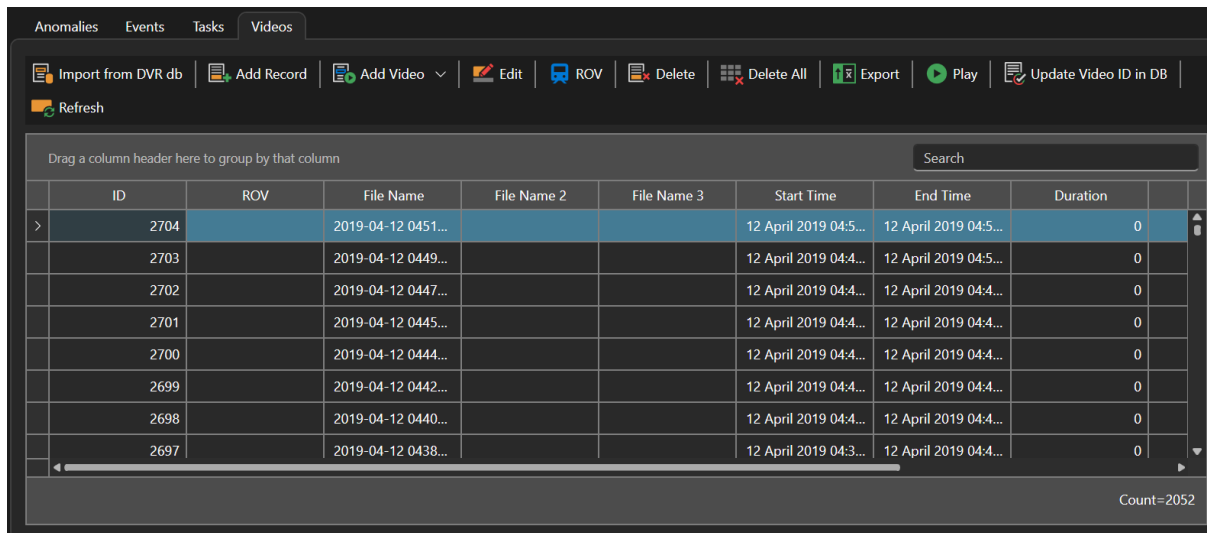


Figure 12. Video records and media commands

- Use Add Video to register an approved media file and Add Record when creating a database record without importing media.
- Confirm File Name, Dive No., ROV, Start Time, and End Time before using the record.
- Use Play for review and Offline Review when media must be reviewed without the live inspection workflow.
- Use Import from DVR db only for the supported source format selected in the interface (for example, MariSoft, Nexus, Eiva, or VisualSoft).
- Use Update Video ID in DB only after confirming the target database and video set; wait for the update operation to finish.
- Use Delete or Delete All only with explicit authorization and after confirming that required evidence is preserved.

Media paths: If video does not open, first verify that the file path is accessible. Reconnecting with the correct Video Folder is safer than editing many records individually.

7.1 Remote DVR connection and logging

Use Remote DVR when the inspection workflow is configured to exchange commands and logging information with an approved DVR endpoint.

The screenshot shows the 'Remote DVR' application window with the following sections:

- Remote Application:**
 - DVR IP Address: 192.168.50.222
 - Local IP Address: 192.168.50.222
 - Send Command Port: 5444
 - ☒ Start listening when opened
- Receive Ports:**
 - Survey String: 44448
 - DVR Status: 44449
 - Receive Command: 44447
- Video Logging:**
 - ROV Name: Shark G3R (dropdown)
 - Dive No.: 2
 - Comment: 2323
 - DVR Online Images: C:\Users\haith\OneDrive\Documents\MariSoft_Videos\Online_Images (file explorer icon)
 - DVR Logging Path: ZULF
 -
- Controller:**
 - (red circle icon)
 - (blue square icon)
 - (split icon)
- Status:**
 - Last Command: Refreshed DVR connection at 04:51:31.
 - DVR Status: (empty field)
 - Survey String: (red bar)
 - Received from DVR: (empty field)

At the bottom, there are five buttons: Start Listening, Stop Listening, Set Default, Save, and Close.

Figure 13. Remote DVR connection, logging, and controller window

- Verify the DVR IP address, local IP address, send-command port, and receive ports against the approved configuration.
- Select the correct ROV and Dive No., then confirm the online-image and DVR logging paths before sending a path or starting logging.
- Use Start listening when receiving DVR messages is required. Check the Status area for the last command, DVR status, survey string, and received data.
- Use Start and Stop to control the configured logging operation. Use Split only when the active workflow requires a file or recording split.
- Choose Save or Set Default only when authorized to change the stored Remote DVR configuration.

Network safety: Do not change IP addresses, ports, logging paths, or default settings during an active inspection unless the approved DVR/network procedure requires the change.

8. Record events

An event is an inspection observation that may later require escalation to an anomaly.

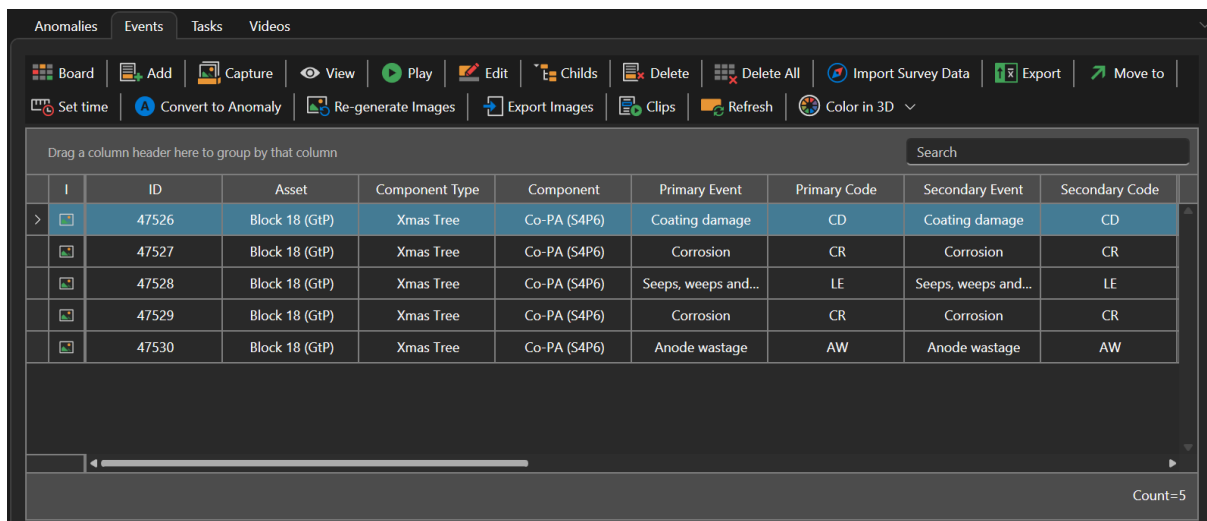


Figure 14. Events tab and event-management commands

- 1 Select the component** Choose the correct tree node and task context.
- 2 Add the event** Choose Add, then complete the applicable primary/secondary codes, comments, water depth, event date/time, and remedial action fields.
- 3 Add evidence** Use Capture, Export Images, Clips, or Import Survey Data as required. Confirm images and clips show the intended observation.
- 4 Review and save** Check the component, codes, time, and description before closing the edit form.
- 5 Escalate if required** Choose Convert to Anomaly only after review confirms the event requires anomaly tracking.

Conversion: Converting an event creates linked anomaly data. Confirm the selected event and answer the confirmation prompt carefully.

9. Manage anomalies and actions

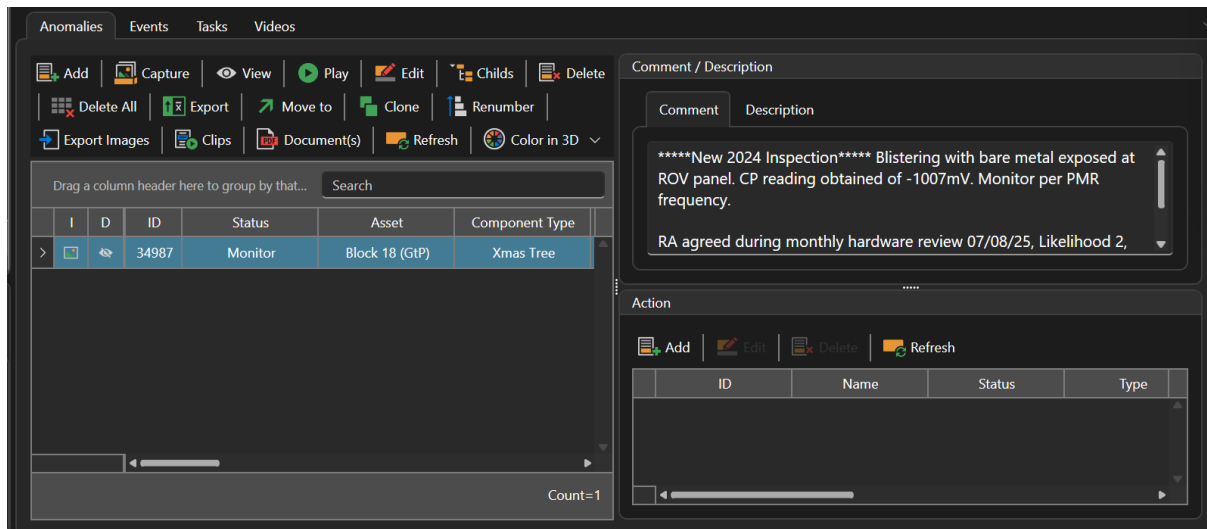


Figure 15. Anomaly record, comments, and actions

9.1 Create or edit an anomaly

- 1 Open Anomalies** Select the correct component and anomaly record, or choose Add.
- 2 Complete the record** Enter or verify code, description/comments, water depth, severity, risk, likelihood, status, year, and reference information as applicable.
- 3 Attach supporting material** Use Capture, Export Images, Clips, and Document(s) to collect or review evidence.
- 4 Set status deliberately** Use the status that matches the approved process (for example Open, Monitor, Referred, Remediate, or Concluded). Do not close an anomaly solely because media review is finished.
- 5 Refresh and verify** Confirm the saved anomaly remains attached to the correct component and workpack.

9.2 Record actions

- Select the anomaly, then use Action to add or review follow-up activity.
- Record a clear action description, responsible person or role when available, creation date, completion date, and status.
- Use Action Before when the action must be completed by a defined date.
- Mark an action complete only when the required evidence or confirmation is available.

9.3 High-impact anomaly commands

Command	Use	Required check
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Command	Use	Required check
Clone	Create a similar anomaly record.	Change all copied fields that do not apply; confirm component and evidence links.
Move to	Associate the record with another valid destination.	Confirm destination component/workpack and downstream reporting impact.
Renumber	Regenerate anomaly name/reference using the configured code-number-year rule.	Use only under the approved numbering process; record the reason if local governance requires it.
Delete / Delete All	Remove eligible anomaly records.	Confirm the selected records and that related evidence or actions will not be lost.

Destructive operations: Delete and bulk-delete commands display confirmation prompts. Read the prompt, confirm the record count, and select No if anything is unexpected.

10. Save the workspace, back up, and export

10.1 Workspace layout

- Workspace > Save Workspace stores the current pane/layout arrangement.
- Workspace > Load Workspace restores a saved arrangement.
- Workspace > Reset Workspace returns to the standard layout when panes are missing or unusable.
- Theme > Apply Dark Theme or Apply Light Theme changes appearance only; it does not change inspection data.

10.2 Database backup

- 1 Confirm the database** Verify the active database in the status bar.
- 2 Choose Backup Database** From the Database menu, select the backup command and choose the approved destination.
- 3 Wait for completion** Do not close the application or disconnect storage while the backup is running.
- 4 Verify the result** Confirm the success message and that the backup file exists in the intended folder.

10.3 Copy, clone, and export

- Copy Workpack (All Data) copies selected components, tasks, and selected related data between databases. Review source, destination, selected workpack, and options before Copy.
- Clone Workpack (Empty Tasks) creates the workpack/task structure without completed inspection data; use only for approved preparation workflows.

- Export to Viewer packages data for the supported viewer workflow.
- Export to Nexus (Excel Sheets) creates an Excel workbook for the chosen survey set, task types, components, and task data. If selected, task images are placed in an Exported_Images folder beside the workbook.
- Do not move or rename an exported workbook independently from its image folder when the workbook contains relative image references.

Before any copy/export: Confirm the active source database and workpack, the destination path/database, the selected records, and whether images, videos, findings, signoffs, anomalies, actions, and alarms are included.

11. Troubleshooting

Symptom	What to check	Next action
Inspection is disabled	A database has not been loaded, or connection failed.	Return to Database Connection, verify selections, and choose Load Data.
Expected workpack is missing	Region, Site, Asset, or Schedule may be wrong; the workpack may not exist in this database.	Recheck selections in order. Contact the workpack owner if still missing.
Grid is empty	Wrong tree node, filter/search text, or context; data may not exist.	Clear filters, select the intended node, and use Refresh.
Video will not play	File path, video folder, file availability, or codec/media support.	Verify access and reconnect with the approved Video Folder; contact support if the file itself fails.
3D/point cloud is blank	The selected component may lack a valid link, or the source file is unavailable.	Verify the model is open, select the correct node, and check the link/path.
Cannot edit a record	No row/node selected, command disabled, insufficient role, or record state prevents editing.	Select the record and retry; otherwise contact the system administrator.
Export appears stuck	Large media sets can take time; a busy indicator may still be active.	Wait for completion. Do not start a second export or close the application.
Unexpected error message	Read the full message and note what action preceded	Stop repeated attempts;

Symptom	What to check	Next action
	it.	capture the message, database/workpack context, and steps for support.

12. Session checklists

12.1 Start-of-session

- Open the approved job folder.
- Connect to the approved database file or SQL Server database.
- Select the correct Region, Site, Asset, Schedule, and Workpack.
- Load Data and verify every context value in the status bar.
- Confirm the required video/model/document paths are available.
- Select the correct component before starting work.

12.2 End-of-session

- End completed tasks and verify start/end times.
- Review that notes, events, anomalies, images, clips, and documents are linked to the correct records.
- Refresh key grids and confirm saved statuses.
- Complete the required backup or export and verify its destination.
- Ensure no copy, export, video-ID update, or media operation is still running.
- Record or hand over any incomplete task or unresolved error according to local procedure.

13. Support information to provide

When reporting a problem, provide enough context for support to reproduce it without exposing credentials:

- Application version shown in the title bar.
- Database name and whether the connection is a database file or SQL Server (do not send passwords).
- Asset, Schedule, Workpack, selected component, and record ID where applicable.
- Exact command used and the sequence immediately before the problem.
- Full error text and a screenshot, with sensitive information removed.
- Whether the problem persists after Refresh or after reconnecting to the same approved context.

Data protection: Share database files, exports, inspection media, and screenshots only through approved

channels and only with authorized recipients.